

Risky Business Cyle

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Can ex ante changes in required excess returns be a major driver of macroeconomic fluctuations?

1. Empirical analysis:

- Identify the shock that explains most of the variance of risk premia.
- Show that this shock (i) generates a business cycle like response in the economy (ii) explain a big portion of the volatility of macro-aggregates.

2. Propose a novel real model where risk premia fluctuations generates business cycle co-movements.

- Risk premia fluctuations: shocks to risk aversion.
- Core idea: reallocate savings from more productive store of value to less productive and safer stores of value.
- Labor relationships play a role as store of value.

Empirical Analysis

Data from 1985Q1 to 2018Q4.

Run a VAR with:

$$Y_t = [gdp_t, c_t, inv_t, n_t, r_t^s, r_t^b, dp_t]'$$

r_t^s : real ex-post stock returns (inclusive of dividends)

r_t^b : real ex-post three-month treasury bill returns

dp_t : dividend price ratio

And the auxiliary Regression:

$$S_t = \Gamma(L)Y_t + \nu_t$$

S_t : Part-time employment; Ex-post return on five-year US bond;
Spread between corporate debt and the 10-year treasury yield.

Empirical Analysis

The Core VAR specification:

$$Y_t = B(L)Y_{t-1} + A\epsilon_t$$

Define the realized j -period accumulated excess return as:

$$rp_{t,t+j} = [r_{t+1}^s + r_{t+2}^s + \dots + r_{t+j}^s] - [r_{t+1}^b + r_{t+2}^b + \dots + r_{t+j}^b]$$

Choice of j : 20 Q (5 years)

Then the VAR implied expectation (for the case with one lag):

$$E_t[rp_{t,t+1}] = (e_5 - e_6)(B + B^2 + \dots + B^j)(I - BL)^{-1}A\epsilon_t$$

Max-share identification strategy, Uhlig (2003).

- Pick A so that the first shock explains most of the volatility of $E_t[rp_{t,t+1}]$.

Empirical Analysis - Expected Excess Return

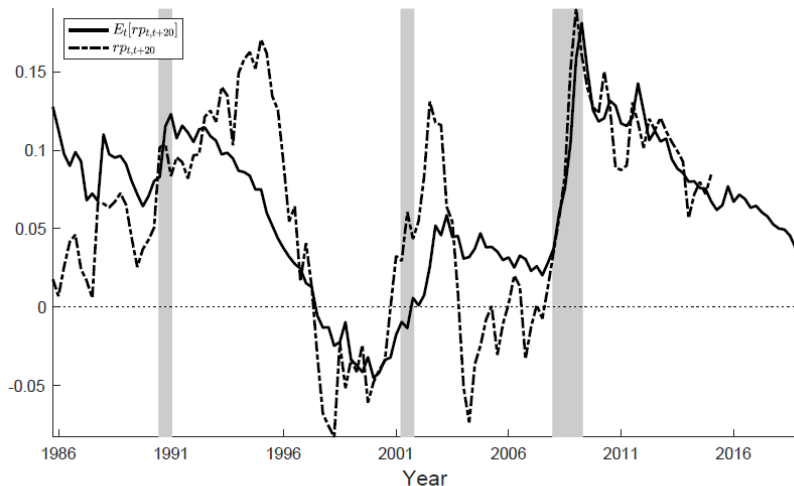
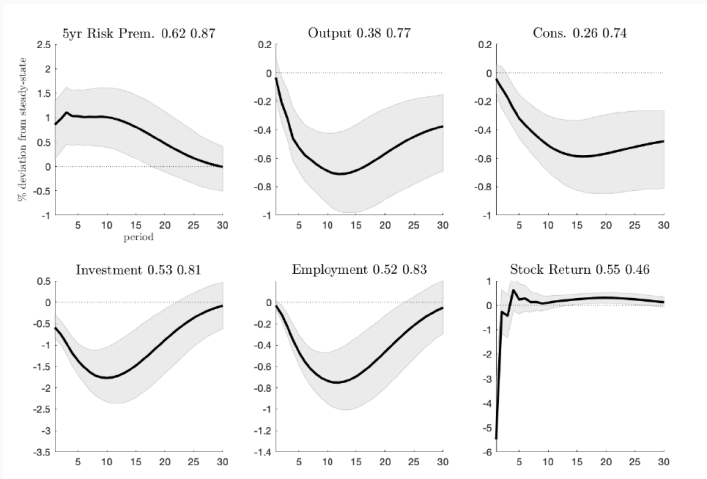


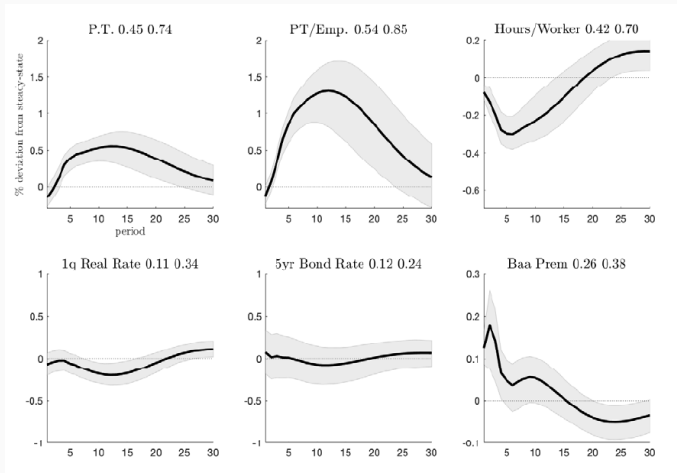
Figure 1: Ex-ante and ex-post excess stock returns.

Empirical Analysis - IRF risk premium shock



- Generates standard business cycle comovements.
- Explain a lot of the volatility of Y, C, I and employment.

Empirical Analysis - Auxiliary Variables



- Employment in Part time jobs increases with the shock.

Model - Overview

Model of a Real Economy with:

- Epstein Zin preferences
- Full time and part time employment.
 - Sticky Wages vs Flexible wages
- Friction in forming and severing labor relations.
- Variable capacity utilization
- Capital adjustment cost

Challenge: generate comovement between C_t and I_t with a shock to risk aversion.

- Risk aversion $\Rightarrow$ Savings reallocated from more risky and productive Full time employment to other assets.
- $\downarrow Y_t \Rightarrow \downarrow C_t$ and $\downarrow I_t$
- $\downarrow N_t \Rightarrow \downarrow MPK \Rightarrow \downarrow I_t$

Model - Households

$$V_t = \max_{C_t, X_{t+1}, B_t, B_t^c} \left[(1 - \beta) C_t^{1 - \frac{1}{\psi}} + \beta (E_t V_{t+1}^{1 - \gamma_t})^{\frac{1 - 1/\psi}{1 - \gamma_t}} \right]^{\frac{1}{1 - 1/\psi}}$$

s.t.

$$C_t + P_t^e X_{t+1} + Q_t^c (B_{t+1}^c - dB_t^c) + \frac{1}{R_t^r} B_{t+1} \leq (D_t^e + P_t^e) X_t + B_t^c + B_t + E_t^l$$

X: Firms equity.

B: Riskless 1 period bond. (zero net supply)

B^c: LT corporate bond - For each unit of B^c , receive 1 and carry dB^c to the future, i.e. payment profile: $(1, d, d^2, d^3, \dots)$.

E^l: revenue from labor:

$$E_t^l = W_{1,t} N_{1,t} + (W_{2,t} + \kappa_t) N_{2,t} + b_{2,t} (1 - N_{1,t} - N_{2,t})$$

Model - Firms

Firm maximizes expected discounted cash flow:

$$E_t \sum_{s=0}^{\infty} \left(\frac{\partial V_t / \partial C_{t+1}}{\partial V_t / \partial C_t} \right) D_{t+s}$$

Subject to:

$$D_t = Y_t - W_{1,t}N_{1,t} - W_{2,t}N_{2,t} - I_t - \gamma_{1,t}^l v_{1,t} - \gamma_{2,t}^l v_{2,t}$$

$$Y_t = (u_t K_t)^\alpha (Z_t N_t)^{1-\alpha}$$

$$K_{t+1} = \left(1 - \delta(u_t) - \frac{\phi_K}{2} \left(\frac{I_t}{K_t} - \delta \right)^2 \right) K_t + I_t$$

$$\delta(u_t) = \delta + \delta_1(u_t - u) + \frac{\delta_2}{2}(u_t - u)^2$$

Leverage is exogenous:

$$Q_t^c L_t = \xi K_{t+1} \quad \text{and} \quad B_{t+1}^c = dB_t^c + L_t$$

Cash flow: $D_t^E = D_t - B_t^c + \xi K_{t+1}$

$$N_t = \left((1 - \Omega) N_{1,t}^{\frac{\theta-1}{\theta}} + \Omega N_{2,t}^{\frac{\theta-1}{\theta}} \right)^{\frac{\theta}{\theta-1}}$$

Firm post full time and part time vacancies: $v_{1,t}, v_{2,t}$ at the cost $\gamma_{1,t}^l, \gamma_{2,t}^l$.

$$N_{i,t} = (1 - \rho_i) N_{i,t-1} + \Theta_{i,t} v_{i,t}$$

$\Theta_{i,t}$: proportion of vacancies i filled, given by the matching function:

$$\mathcal{M}_i = \chi_i \nu_i^{\epsilon_1} S_i^{1-\epsilon_i}$$

$$\Theta_{i,t} = \chi_i \left(\frac{\nu_{i,t}}{S_{i,t}} \right)^{\epsilon_i - 1}$$

Workers search sequentially.

$$S_{1,t} = 1 - (1 - \rho_1) N_{1,t-1} - (1 - \rho_2) N_{2,t-1}$$

$$S_{2,t} = 1 - N_{1,t} - (1 - \rho_2) N_{2,t-1}$$

Model - Labor Market

- Full time workers, N_1 : Nearly fixed wages, $W_{1,t}$
- Part time workers, N_2 : Nash-bargain wage, $W_{2,t}$

Value of a math for the firm:

$$J_{i,t} = MPL_{i,t} - W_{i,t} + (1 - \rho_i)E_t[M_{t,t+1}J_{i,t+1}]$$

Or

$$J_{i,t} = \sum_{j=0}^{\infty} (1 - \rho_i)^j \frac{E_t(MPL_{i,t+j} - W_{i,t+j})}{R_t^r} + \\ + \sum_{j=0}^{\infty} (1 - \rho_i)^j Cov_t(M_{t,t+1}, MPL_{i,t+j} - W_{i,t+j})$$

Model - Closing

Z_t is integrated of order one:

$$\log(Z_t) = \log(Z_{t-1}) + \sigma_z \epsilon_t^z$$

Risk aversion dynamics:

$$\log(\gamma_t/\gamma_{ss}) = \rho_{1,\gamma} \log(\gamma_{t-1}/\gamma_{ss}) + \rho_{2,\gamma} \log(\gamma_{t-2}/\gamma_{ss}) + \sigma_\gamma \epsilon_t^\gamma$$

Cointegration of $X \in \{\gamma_{1,t}^l, \gamma_{2,t}^l, W_{1,t}, b_{2,t}, G_t\}$ with technology:

$$X_t = \Gamma_t \bar{X}$$

Where $\bar{X}$ is the steady state value of X and Γ_t follows:

$$\Gamma_{t+1} = \Gamma_t^\omega Z_t^{1-\omega}$$

ω - How fast the Γ process tracks productivity.

Quantifying the mechanism

Quantifying the mechanism

Table 3: Calibrated Parameters

Name	Description	Value
β	Discount rate	0.994
ψ	Intertemporal elasticity of substitution	2.500
α	Capital share	0.300
δ	Capital depreciation rate	0.025
δ_2	Cap. util. cost $\times 100$	0.030
$\bar{g}$	Steady-state G/Y	0.200
d	Corporate bond duration	0.975
Labor Markets		
ρ_1	Separation Rate - FT	0.044
ρ_2	Separation Rate - PT	0.354
η_1	HH's bargaining power - FT	0.500
η_2	HH's bargaining power - PT	0.200
Exogenous Processes		
σ_z	Std. dev. of tech shock	0.006

- $\rho_1/\rho_2 = 8$ matches difference in separation rates.
- ρ_1 matches overall separation rate (10%).
- $\rho_z = 0$

Quantifying the mechanism - Estimation

- Solve model using 3rd-order perturbation
- Target impulse response of $\{Y_t, C_t, N_t, N_{2,t}, r_t^s, r_t^b\}$
- Also target unconditional moments

Table 4: Unconditional Target Moments

Description	Data	Model
Equity risk premium	0.064	0.063
Share of part-time	0.180	0.182
LR unemployment	0.060	0.061
Vacancy Rate	0.035	0.035
Hiring cost/GDP	0.010	0.010
PT earn./FT earn.	0.500	0.497
Std. Dev. HP log(Emp/Pop)	0.011	0.011
Std. Dev. HP log(vacan.)	0.117	0.117

Quantifying the mechanism - Estimation

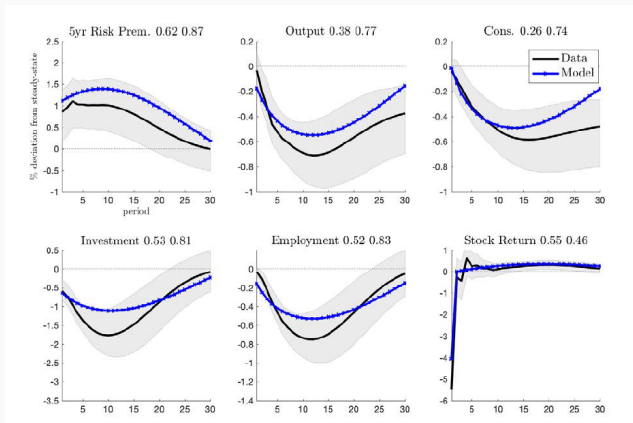
Table 5: Estimated Parameters

Name	Description	Point Est.	Std Err.
γ_{ss}	Steady-state risk aversion	79.123	11.508
ϕ_k	Capital Adj. Cost	12.478	2.210
ν	Leverage Ratio	0.822	0.024
Labor Markets			
γ_1^l	Vacancy posting cost - FT	4.092	0.442
γ_2^l	Vacancy posting cost - PT	0.144	0.020
b_1	Value if no perm posit.	1.108	0.045
b_2	Value if unemployed	0.645	0.006
Ω	Labor contrib. of PT	0.172	0.008
θ	Elas. between FT & PT	1.715	0.108
ϵ_1	Matching elasticity - FT	0.329	0.023
ϵ_2	Matching elasticity - PT	0.514	0.016
χ_1	Matching technology - FT	0.794	0.051
χ_2	Matching technology - PT	0.959	0.020
ω	Cointegration parameter (i.e. wage adj.)	0.968	0.009
Risk Aversion Process			
$\rho_{1,\gamma}$	AR(1) risk av. shock	1.960	0.026
$\rho_{2,\gamma}$	AR(2) risk av. shock	-0.966	0.025
σ_γ	Std. dev. of risk av. shock	0.025	0.010

$$\nu = \frac{\xi}{(1-d)}$$

Quantifying the mechanism - Estimation

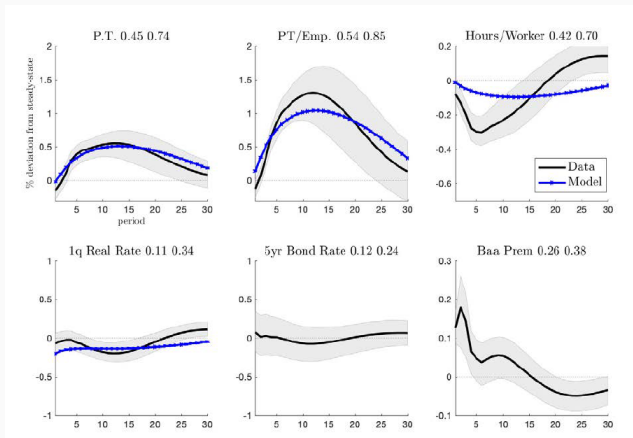
A Risk Premium Shock



- Shift from FT to safer PT employment $\Rightarrow \downarrow Y_t$
- Reduced MPK $\Rightarrow$ reduced incentives to invest.

Quantifying the mechanism - Estimation

A Risk Premium Shock

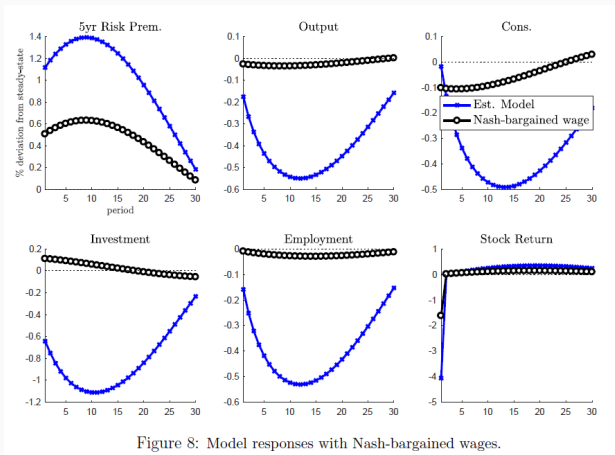


- Model matches well the labor reallocation.
- Small interest rate change. Fluctuation driven by risk premium.

Inspecting the Mechanism

Counterfactual: Flexible Wages

Turning off sticky wages.



- Reallocation to less productive employment largely absent.
- Output nearly unaffected and investment rises.

Counterfactual: No Risk Premium

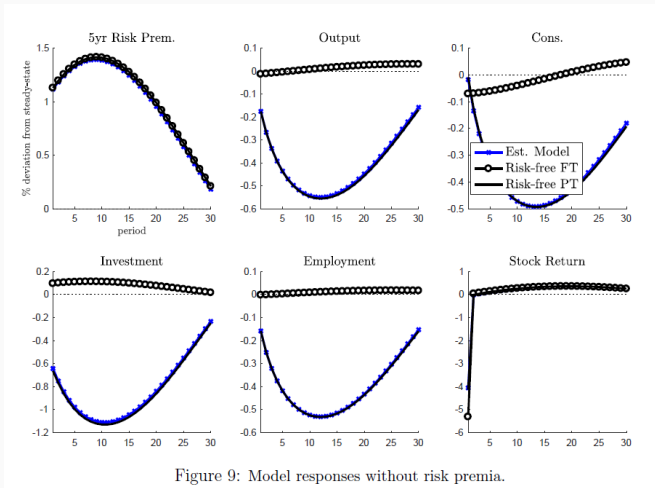
Erasing Risk Premium.

$$J_{i,t} = \sum_{j=0}^{\infty} (1 - \rho_i)^j \frac{E_t(MPL_{i,t+j} - W_{i,t+j})}{R_t^r} +$$

~~$$+ \sum_{j=0}^{\infty} (1 - \rho_i)^j \text{Cov}_t(M_{t,t+1}, MPL_{i,t+j} - W_{i,t+j})$$~~

No risk premium, labor is safe.

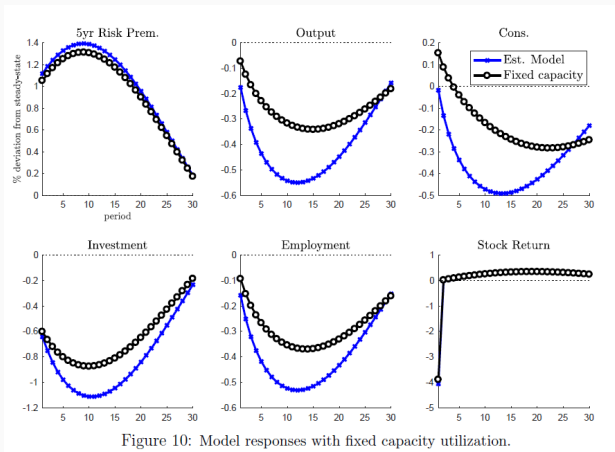
Counterfactual: No Risk Premium



- Riskiness of FT employment drives the result.
- Variation in the risk premia of matches, not volatility of the expected cash flow.

Counterfactual: Fixed Capacity Utilization

Fix capacity utilization.



- Inv. still falls due to $\downarrow N_t$ and $\downarrow MPK$
- Output fall not as big, so $\uparrow C_t$

Conclusion

Risk-premia changes are strongly associated with macroeconomic fluctuations.

Theory where causality flows from premia $\Rightarrow$ macro fluctuations

- risk premia drive reallocation of saving towards safer but less productive stores of value.
- Emphasis on labor market guided by the empirical results

Captures many qualitative and quantitative features of the cycle.